

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T37 · Transformations of Graphs

Questions

Student book · questions with writing space

■ 26 pages

FAMILY MAP

Transformations of Graphs

F01 Map a known graph feature

17 Q | 28 marks

F02 Infer or describe the graph transformation

3 Q | 7 marks

F03 Write the transformed equation

5 Q | 10 marks

19 The equation of a curve is $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are $(8, -12)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 3$

(.....,)
(1)

(ii) $y = f(2x)$

(.....,)
(1)

(Total for Question 19 is 2 marks)

Question 09

4MA1-1H Q21

MODULAR UNIT 2

2 marks

21 A curve has equation $y = f(x)$

There is one minimum point on this curve.

The coordinates of this minimum point are $(5, -4)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 7)$

(..... ,)
(1)

(ii) $y = f(x) - 6$

(..... ,)
(1)

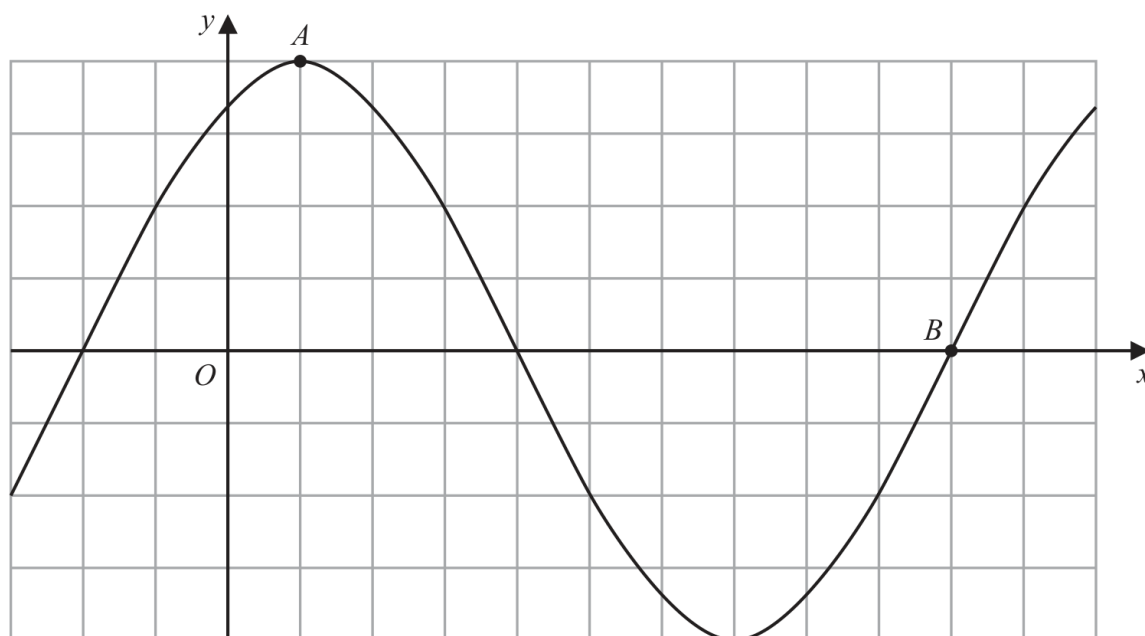
Question 27

4MA1-2H Q25

MODULAR UNIT 2

2 marks

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)^\circ$



(i) Find the coordinates of the point A

(.....,)
(1)

(ii) Find the coordinates of the point B

(.....,)
(1)

Question 19

4MA1-1H Q24 (a)

MODULAR UNIT 2

2 marks

24 The curve with equation $f(x) = 5x^2 + 9x + 2$ is transformed to the curve with equation

$$g(x) = 5(x+4)^2 + 9(x+4) + 8 \text{ by the translation } \begin{pmatrix} a \\ b \end{pmatrix}$$

(a) Write down the value of a and the value of b

 $a = \dots\dots\dots$ $b = \dots\dots\dots$

(2)

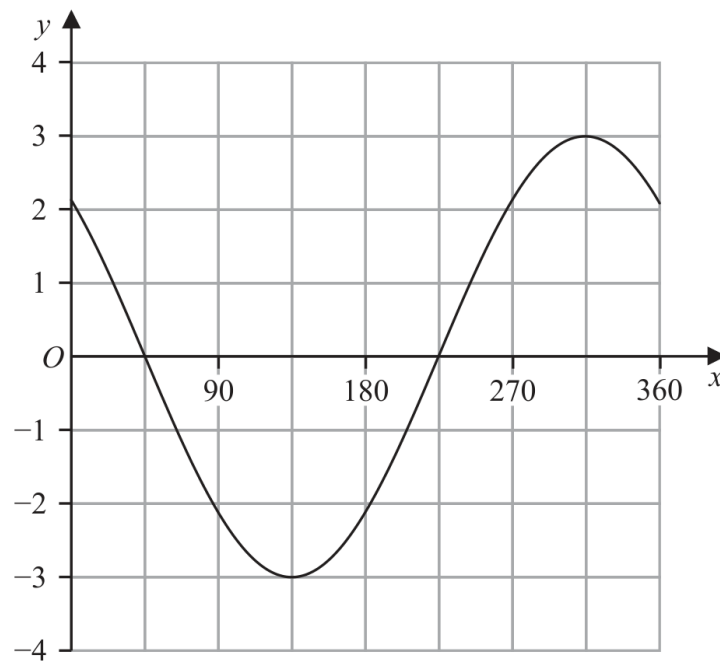
Question 20

4MA1-1H Q24 (b)

MODULAR UNIT 2

2 marks

The graph of $y = p \cos(x + q)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid below.



Given that $p > 0$ and $0 < q < 360$

(b) find the value of p and the value of q

$p = \dots\dots\dots$

$q = \dots\dots\dots$

(2)

Question 17

4MA1-1H Q21

MODULAR UNIT 2

2 marks

21 A curve has equation $y = f(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (6, 5)

Write down the coordinates of the turning point on the curve with equation

(a) $y = f(x - 4)$

(.....,)
(1)

(b) $y = f(3x)$

(.....,)
(1)

Question 30

4MA1-2H Q19 M01

MODULAR UNIT 2

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are (5, 4)

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 5)$

(.....,)
(1)

Question 31

4MA1-2H Q19 M02

MODULAR UNIT 2

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are (5, 4)

(ii) $y = 3f(x)$

(.....,)
(1)

Question 32

4MA1-2H Q19 M03

MODULAR UNIT 2

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are (5, 4)

(iii) $y = f(x) - 7$

(.....,)
(1)

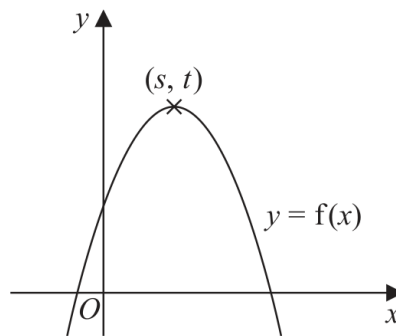
Question 16

4MA1-1H Q20 Q20(i)

MODULAR UNIT 2

1 marks

20 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.

The coordinates of this maximum point are (s, t)

Find, in terms of s and t , the coordinates of the maximum point on the curve with equation

(i) $y = f(x - 2)$

(.....,)
(1)

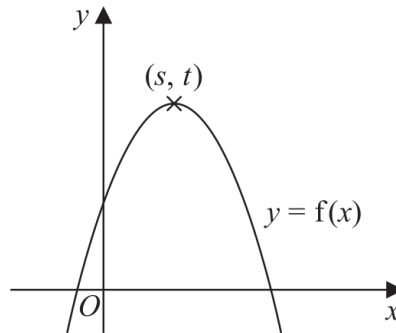
Question 17

4MA1-1H Q20 Q20(ii)

MODULAR UNIT 2

1 marks

20 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.

The coordinates of this maximum point are (s, t)

Find, in terms of s and t , the coordinates of the maximum point on the curve with equation

(ii) $y = 3f(x)$

(.....,)

(1)

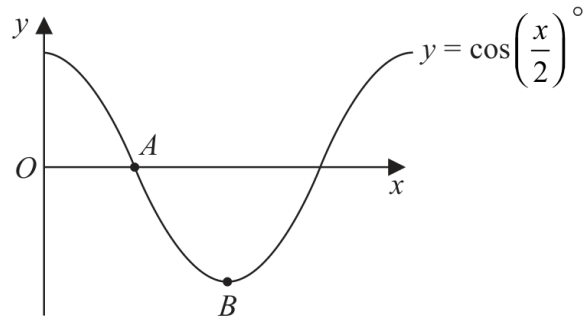
Question 31

4MA1-2H Q23 Q23(i)

MODULAR UNIT 2

1 marks

- 23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



- (i) Find the coordinates of the point A

(.....,)
(1)

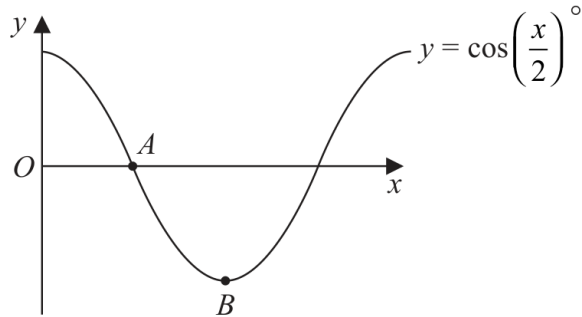
Question 32

4MA1-2H Q23 Q23(ii)

MODULAR UNIT 2

1 marks

- 23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



- (ii) Find the coordinates of the point B

(.....,)

(1)

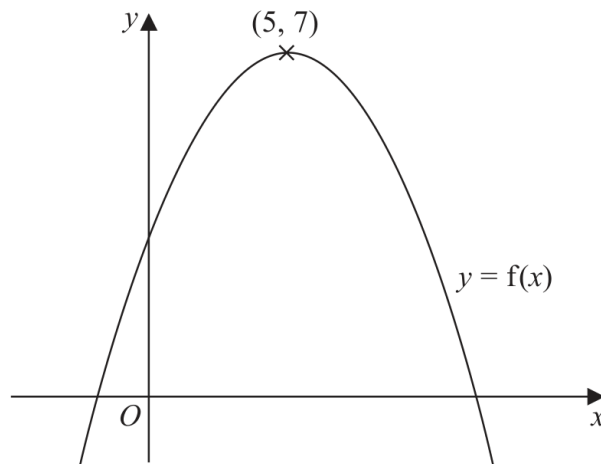
Question 14

4MA1-1H Q23

MODULAR UNIT 2

2 marks

23 The diagram shows a sketch of the curve with equation $y = f(x)$



There is only one maximum point on the curve.
The coordinates of this maximum point are (5, 7)

Write down the coordinates of the maximum point on the curve with equation

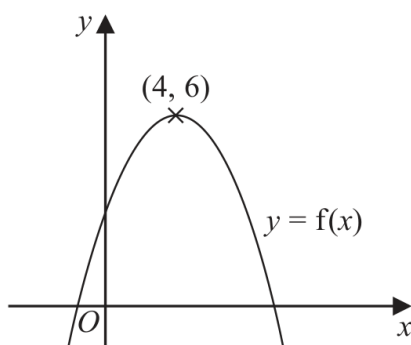
(i) $y = f(x + 9)$

(.....,)

(ii) $y = f(x) + 3$

(.....,)

21 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.

The coordinates of this maximum point are (4, 6)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 4)$

(.....,)

(ii) $y = f(2x)$

(.....,)

(2)

The curve **C** is transformed to curve **S** under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$

You do not need to simplify the equation.

The coordinates of the minimum point on this curve are $(-9, 15)$

(i) $y = f(x + 3)$

(.....,)

(ii) $y = \frac{1}{3} f(x)$

(b) find the value of a and the value of b .

$$b = \dots\dots\dots$$

By writing $g(x)$ in the form $a(x - b)^2 - c$, where a , b and c are constants, describe fully a series of transformations that map the curve **S** onto the curve **T**.

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

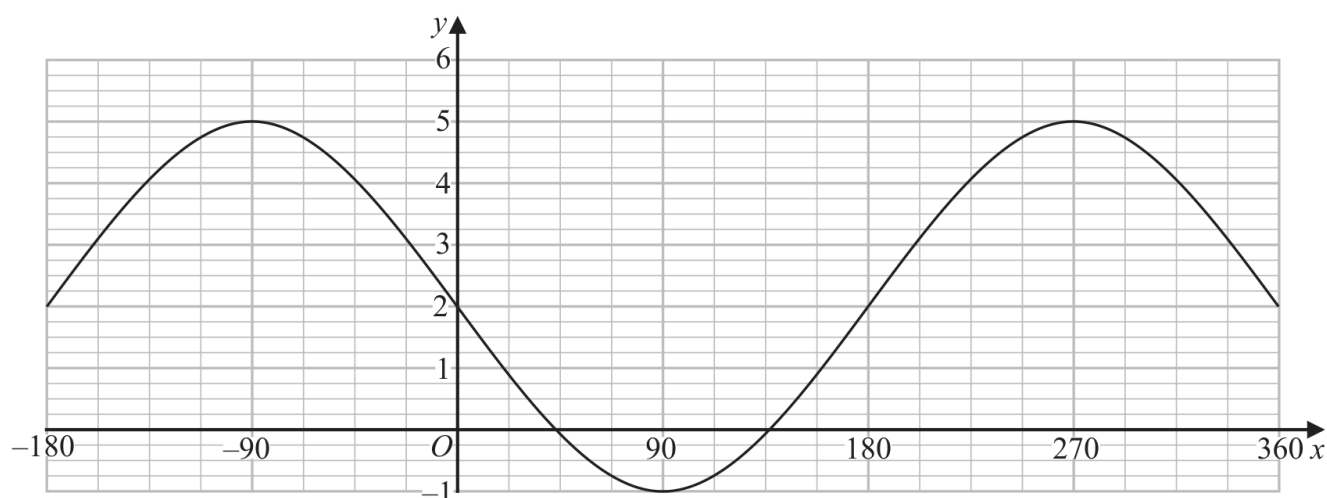
.....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

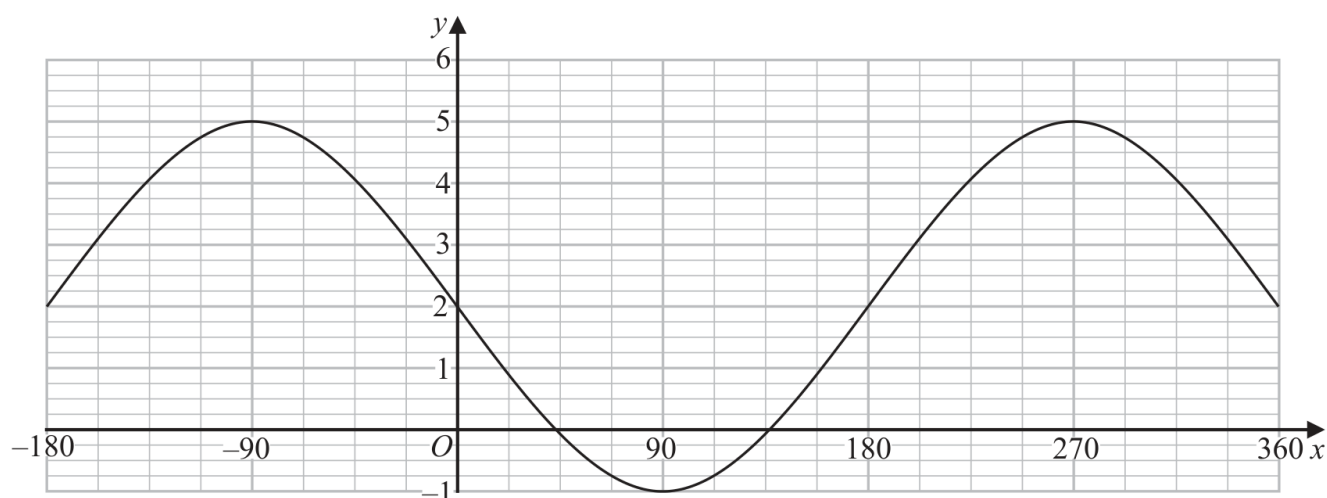
.....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

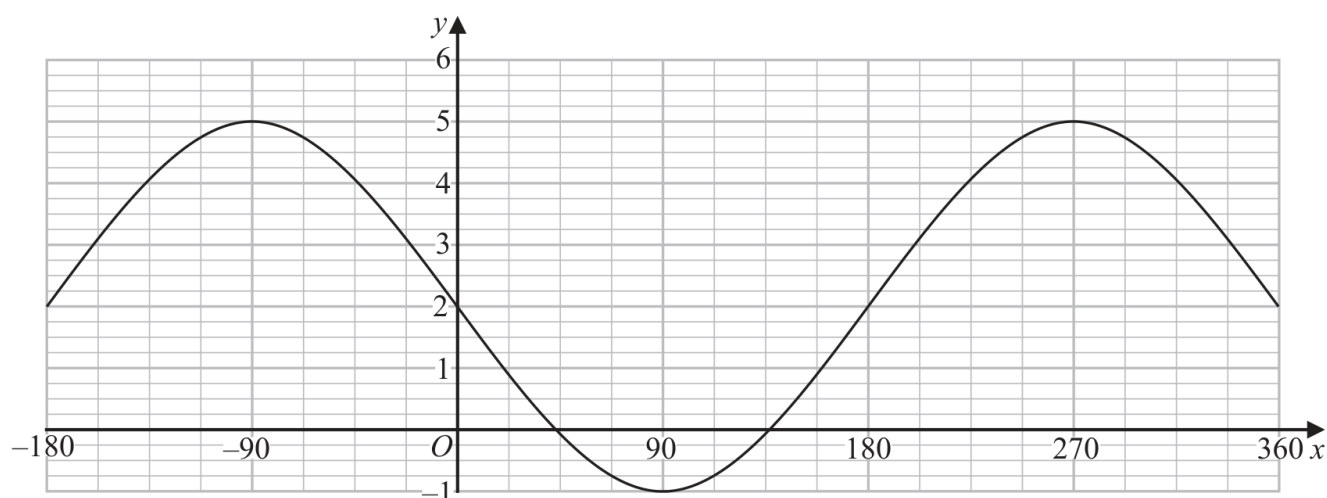
.....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

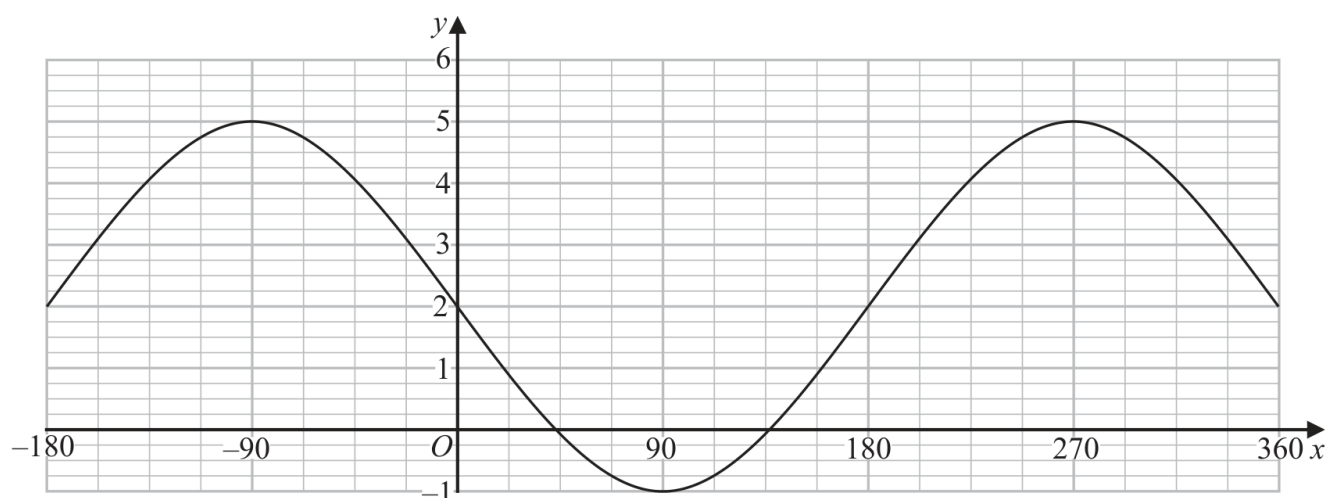
.....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



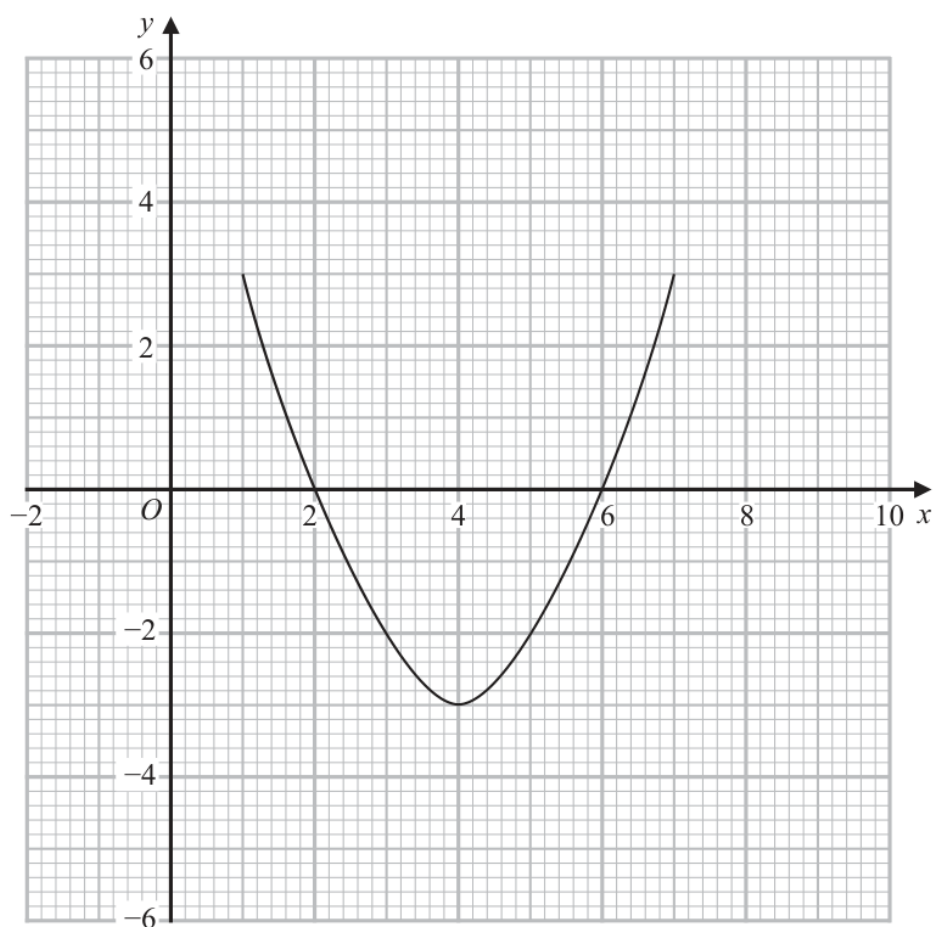
(d) Find the value of a , the value of b and the value of c .

- 25** The curve with equation $y = g(x)$ is transformed to the curve with equation $y = -g(x)$ by the single transformation **T**.

(a) Describe fully the transformation **T**.

(1)

The diagram shows the graph of $y = f(x)$



(b) On the grid, draw the graph of $y = 2f(x - 1)$

(2)

(Total for Question 25 is 3 marks)

24 A curve has equation $y = f(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (4, 3)

Write down the coordinates of the turning point on the curve with equation

(i) $y = f(x + 5)$

(..... ,)
(1)

(ii) $y = f(x) + 7$

(..... ,)
(1)

(iii) $y = f(2x)$

(..... ,)
(1)

(Total for Question 24 is 3 marks)